**Supplementary material – ESM 1 – Study Protocol**

Title

Pre-existing diabetic retinopathy as a prognostic factor for COVID-19 outcomes amongst peoples with diabetes: A systematic review

Authors

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## Background

Description of the health condition and context

- The targeted health condition is poor COVID-19 prognosis for people with diabetes mellitus, of any age, gender or ethnicity.
- The prognostic factor under investigation is diabetic retinopathy and must be pre-existing prior to COVID-19 infection.
- The clinical context for which diabetic retinopathy as a potential prognostic factor is intended is the COVID-19 Pandemic ( December 2019 – Current)
- Other reviews investigating prognostic/risk factors for the diabetic population are currently ongoing according to PROSPERO site for COVID-19 reviews. However, none are specific to diabetic retinopathy.

Description of the prognostic factor

- Pre-existing diabetic retinopathy is defined as a diagnosis of retinopathy prior to COVID-19 infection. Diagnosis can be made via diabetic eye screening.
- The type of studies of interest are prognostic factor studies.
- Any grade of retinopathy is of interest, whether background, pre-proliferative, proliferative (as defined by the WHO for retinopathy).

Health Outcomes

- COVID-19 poor prognosis is defined as any one of: critical care, mortality, hospital admission, intubation, mechanical ventilation, increased hospital stay, sepsis, pneumonia, artificial respiration, ICU admittance, or any other COVID-19 related adverse complication.
- The time horizon for the occurrence of these outcomes was not defined. Therefore, any time horizon was included in the review.

Why is it important to do this Review

- Thankfully, most suffer mild symptoms, however some are at risk of severe prognosis including ICU admission and Death.
- Risk groups have been identified as the elderly, immunocompromised and those with certain comorbidities.
- One such comorbidity is Diabetes Mellitus (DM). DM is globally widespread and on average, appears to double the risk of adverse COVID-19 outcomes. However, DM is a highly heterogenous group with some sharing the risk of the general population while others are at risk of severe prognosis.
- There is a need, therefore, to identify those factors relating to a poorer prognosis in the population with diabetes to aid clinicians with risk stratifying for vaccine prioritisation or for closer monitoring.
- A potential prognostic factor is diabetic retinopathy which is a type of microvascular disease and a common complication of diabetes.
- COVID-19, in its most prolific form, behaves as a microvascular disease and it's been hypothesised that those with pre-existing microvascular disease are more vulnerable.
- Furthermore, as retinopathy can be easily and accessibly imaged, this may provide an efficient diagnostic tool for clinicians.
- This review is timely as increasing literature is highlighting the important role of microvascular diseases and poor covid-19 prognosis.

## Objectives

Primary Objective

- To determine whether pre-existing diabetic retinopathy predicts poorer COVID-19 prognosis in patients with diabetes mellitus.
- Where possible, limitations and gaps in the literature will be identified and future research suggested.

PICOTS table:

| Population | Index prognostic factor | Comparison | Time | Outcome | Setting/ Study Design |
|---|---|---|---|---|---|
| Diabetic population with COVID-19 infection | Pre-existing retinopathy (any grade) | No retinopathy | COVID-19 pandemic<br>The time horizon for outcomes is not defined | Poor COVID-19 prognosis | Non-interventional observational cohort studies<br><br>Setting: patient-specific risk stratification in DM |

Investigations of sources of heterogeneity between studies

- Any sources of heterogeneity between studies will be identified and discussed.

## Methods

Criteria for considering studies for this review:

*Inclusion criteria*

**Study population:**

- People with diabetes infected with COVID-19
    - People with diabetes diagnosed at any point/ had any duration of time with diabetes.
    - any age
    - any gender
    - human
    - evidence of microvascular disease
- subgroups
    - Target population: evidence of pre-existing diabetic microvascular disease (including retinopathy measurement)

- Comparison population: people with diabetes without pre-existing microvascular disease (including retinopathy measurement)

**Intervention of that population:**

- none

**Study type:**

- Non-interventional observational studies
- Monocentric or multicentred
- Observational: cohort (retrospective or prospective)
- Although any observational study can be included, it must measure the relevant outcomes.

**sample size**

- any

**Setting**

- hospital/ICU/medical

**study description**

**Outcome/ endpoint:**

- Main COVID-19 outcome: poor COVID-19 disease prognosis
  - Outcome of poor prognosis is defined as any COVID-19 related poor outcome such as: critical care, mortality, hospital admission, intubation, mechanical ventilation, increased hospital stay, sepsis, pneumonia, artificial respiration, ICU admittance, or any other COVID-19 related adverse complication
- Main endpoint measurement: quantitative measure of severity/prognosis of COVID-19 disease and association to risk factors ie:
  - measuring independent risk factors/ prediction models/ association/ risk stratification scores/ risk scores/ comparison/ phenotypes for poor COVID-19 infection prognosis in the population with diabetes
    - This MUST be a measurement with regards to pre-existing diabetic microvascular disease as an *independent* risk factor for poor COVID-19 prognosis (whereby retinopathy is an included measurement for microvascular disease or independently measured)

**Data type:**

- primary data
- preliminary and final studies
- quantitative
- all studies must report relevant outcome data
- all studies must report data with regards to whether microvascular disease is an independent risk factor (where diabetic retinopathy has been included as a measurement for microvascular disease) with poorer COVID-19 outcomes, such as in the form or OR, RR, comparative sample numbers, predictive model.

**Language:**

- any language

**Geographical location**:

- global

**Time period**:

- COVID-19 pandemic (Dec 2019 – Current)

**Publication date restriction**

- Dec 2019 – June 2021

*Exclusion Criteria*

**Study population: including:**

1. People with diabetes not infected with COVID-19
2. not relating to COVID-19
3. People without diabetes
4. animal studies

subgroups:

- target population: population with diabetes studied does not include a subgroup or total population of patients with pre-existing microvascular disease (including diabetic retinopathy).
- Comparison population: population without diabetes

**Intervention:**

- any

**Study type: including:**

1. non-observational studies: RCTs, editorials, reviews.
2. observational studies not measuring poor COVID-19 prognosis as main COVID-19 outcome.
3. observational studies with no quantitative measure of severity/prognosis of COVID-19 disease and associated risk factors (as defined in the inclusion criteria).
4. observational studies with no measurement of pre-existing microvascular disease (whereby retinopathy is an included measurement for microvascular disease or independently measured).

**Outcome/ Measurement: including:**

5. no measured endpoint with regards to any association between pre-existing diabetic microvascular disease (where diabetic retinopathy included in measurement or independently) and adverse COVID-19 outcomes.
6. Did not measure poor COVID-19 prognosis aw main COVID-19 outcome.

**Data type: including:**

7. no data for the relevant outcomes
8. data collection ongoing
9. data collected doesn't include pre-existing microvascular disease (where diabetic retinopathy included in measurement or independently)
10. no data with regards to how pre-existing diabetic microvascular disease ((where diabetic retinopathy included in measurement or independently) is associated with poorer COVID-19 outcomes.

## Search methods for identification of studies

**Electronic Searches:**

- The search strategy will be based on the review question (including synonyms based off the PICTOS breakdown).
- Initial scope of the literature will help determine what literature is available.
- Relevant identified papers will also be used to develop the literature search, using the Yale MeSH analyser.
- Contact with academic librarians will also be made to help develop the search.

- Chosen databases: Medline, EMBASE, web of science, PubMed, Cochrane COVID-19 Registry.
- The primary search will be adapted to MEDLINE.
- Forward and backward citation searching of relevant papers.
- Database adapted of searches helped with the polyglot tool.
- No language restrictions will be applied.
- No study design restrictions will be included at this stage.

**Searching other sources**

- Chosen grey literature databases: *MedxRiv* and Global Index Medicus
- Other grey literature sources: *Diabetologica, Diabetes UK*

## Data collection

**Selection of studies**

- The pre-determined selection criteria will firstly be applied by the primary author and then by two further qualified reviewers.
- Disagreements will be resolved via online meetings.
- Study selection will be a two-phase process: abstract/title screening followed by full paper screening.
- Any amendments to the selection criteria as a result will be mentioned in the report.

**Data extraction and management**

- Data extraction will be guided by the recommended CHARMS-PF checklist.
- Data extraction will be a dual process, conducted by the primary author and a qualified assessor.

**Assessment of risk of bias in included studies**

- The Quality in Prognostic Studies (QUIPS) tool will be used for risk of bias assessment.
- This will not be a method of study exclusion. All papers included at the screening stage will be included in the review, no matter the quality. However, the risk of bias

assessment will be used to weigh papers with regards to reliability in answering the primary research question.

- This was also dually conducted, first by the primary author and then by a qualified assessor.

**Measures of association to be extracted:**

- Measures of effect size, Odds Ratios (OR) or Hazard Ratios (HR) will be extracted, together with their measure of certainty (95% CI) and their associated p-value.

**Dealing with missing data:**

- Missing data will be identified
- Where possible, contact with authors of the studies with missing data will be attempted for further information.

**Data synthesis:**

- Meta-analysis will only be considered feasible if the included studies are sufficiently comparable. If not, a narrative synthesis of the results will be conducted.
- Any areas and limitations of between-study heterogeneity will be identified and explored in the discussion